



UV-curable boron nitride nanosheet/ionic liquid-based crosslinked composite polymer electrolyte in lithium metal batteries

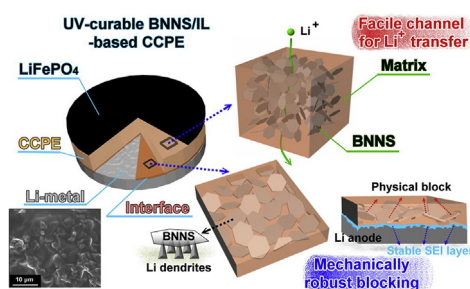
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HIGHLIGHTS

- A new class of crosslinked composite polymer electrolytes is fabricated.
- The rapid and solvent-free curing of the coating film is formed on the electrodes.
- Well-dispersed BNNS and crosslinked structures simultaneously present in the film.
- 3D ion conducting paths is constructed to promote Li-ion transport.
- BNNS is formed a mechanically robust interface to inhibit Li dendrite growth.

GRAPHICAL ABSTRACT



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ABSTRACT

Crosslinked polymer electrolyte (CPE) and composite polymer electrolyte are two of the most promising polymer electrolyte (PE) systems for improving the resistance to Li dendrite growth, but the trade-off and agglomerated effects of these PEs hinder the development of safe Li metal battery (LMB). Herein, a new class of crosslinked composite polymer electrolyte (CCPE) with the combination of confined ionic liquid (IL) polymer network and homogeneously dispersed boron nitride nanosheet (BNNS) for LMB is reported. This UV-curable CCPE is formed on electrode via in-situ crosslinking of a precursor solution in a rapid and solvent-free step, and the assembly process of LMBs and interfacial contact/adhesion are maximum optimized. The critical point of this CCPE is that a facile channel for Li^+ transfer and a robust skeleton for confining IL and strengthening interface are simultaneously achieved via the introduction of network structure and 2D BNNS in PE. Consequently, the $\text{LiFePO}_4/\text{Li}$ batteries involving CCPE show a great improvement in the charge-discharge cycling stability and rate performance compared with those of the cells with pure CPE. This study offers an effective and facile strategy for fabricating dendrite-resistant CCPE with potential applicability in next-generation energy storage systems.

1. Introduction

The Li metal battery (LMB) with extraordinary high theoretical capacity (3860 mAhg^{-1}) and lowest negative redox potential (-3.04 V vs. standard hydrogen electrode) [1], which has considered being one of

the most promising candidates for high energy density batteries in electric vehicles and advanced electronic devices [2]. However, severe safety and poor cycling efficiency caused by the formation of uncontrollable Li dendrite during charge-discharge cycles drags LMBs out of the practical application [3,4]. Therefore, how to inhibit the

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uncontrolled growth of Li dendrites and the resulting “dead Li” have been strongly considered as one of the most critical issues for design-safety and reliable LMBs.

The design of ideal solid electrolyte interphase (SEI) layer on Li metal anode is critical for suppression of the uncontrolled growth of Li dendrites, which should give SEI high Li ionic conductivity to uniform Li metal deposition on the anode and high elastic strength to physical suppress the break through by Li dendrites [5]. Thus, there are extensive methods have been proposed in the literature, such as (i) seeking proper organic solvent as electrolyte [6,7], (ii) employing Li-salts with good comprehensive performance (high safety, low toxicity, wide electrochemical window, etc.) [8], (iii) introducing functional additives to optimize Li surface [9] and (iv) covering an artificial passivation film on the Li surface [10], to build a uniform and stable SEI layer during charge-discharge processes. However, the limited mechanical strength of SEI layers and thermodynamically unstable Li metal in liquid electrolytes are obstacles to their use for suppression of Li dendrite growth. Although solid polymer electrolytes (SPEs) exhibit high mechanical strength and less reactive with Li metal, which are benefit for the suppression in Li dendrite growth [11,12], their performances suffer due to unsatisfactory ionic conductivity and the high interfacial resistance with electrodes. Considering the critical effect of Li-ion conductivity of SPEs on Li dendrites, a large number of electrolyte systems, such as organic solvents [13], ionic liquids (ILs) [14] and succinonitrile (SN) [15,16], have been combined with macromolecular structures for preparing gel PEs (GPEs) or plasticized PEs (PPEs). While all of these systems improve ionic conductivity and interfacial contact/adhesion with the electrodes, but the resulting GPEs and PPEs are easily losing mechanical strength caused by plasticization effect of liquid electrolytes, leading to unstable SEI layer formation and sacrificing resistance to Li dendrite growth. Hence, simultaneously meets all major requirements including high ionic conductivity, good mechanical strength and stable electrode-electrolyte interface of the PEs remains a formidable challenge.

The addition of chemically crosslinked structures [17–19] and inorganic fillers [20–23] to PEs are two considered effective methods of improving the resistance to Li dendrite growth resulting from their improved mechanical properties, thermal and dimensional stabilities. The former strategy involves the thermal/or photo curing of a precursor solution containing polymerizable and cross-linkable monomers, a liquid electrolyte and an initiator (some case is an initiator-free system), which can in-situ synthesize a free-standing thin film in one step thereby simplifying the assembly process of PE-based LMBs. In cross-linked PEs (CPEs), unfortunately, there is a trade-off between their ionic conductivity and mechanical strength, which heavily depends on the crosslinking degree in resulting CPEs. Different from the former one, the latter strategy endows the PEs simultaneously with enhanced ionic conductivity and mechanical strength as well as a stable electrode-electrolyte interface by introduced fillers bearing good dispersion and compatibility. However, the composite PEs is usually a physical blend of fillers, polymer matrix and a conducting salt dissolved in a non-aqueous organic solvent, therefore, non-assisted fillers easily trending to aggregate during the solvent evaporation process, especially when filler contents increase to a certain value. The agglomerated filler domains fail to provide well-defined ion-conducting path and filler network thereby denting the improvements in ion-conductivity and mechanical properties for the resulting composite PEs.

The 2D filler hexagonal boron nitride (BN) or BN nanosheet (BNNS) arises an interesting graphene-analogous for use in the ceramic separator for Li batteries, as a newly developing layered material possessing high thermal conductivity, mechanical properties, being chemically inert and thermally stable [21,24–26]. However, such intrinsic chemically inert surface makes them poor compatibility and dispersibility in most organic solvents and polymers. Although surface functionalization is available for strength their interactive with other media; however, these methods are typically less efficient and involve

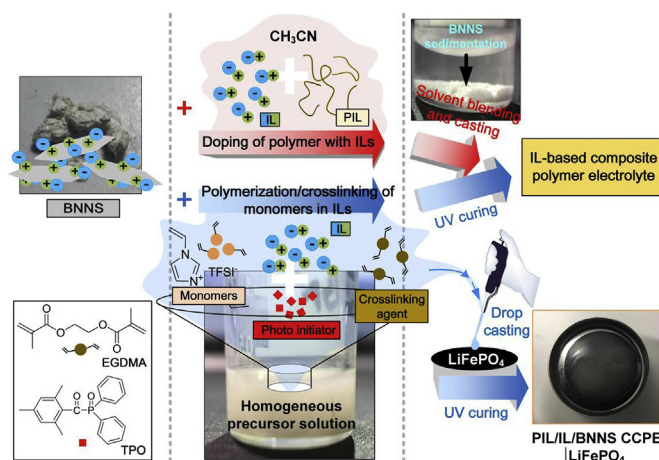


Fig. 1. A schematic representation illustrating the UV-curing and CPE preparation processes. The insert photographs depicting the BNNS, homogeneous precursor solution containing BNNS, monomers, crosslinking agent and photoinitiator, blending of BNNS, IL and polymer ionic liquid (PIL) in CH_3CN and CCPE with BNNS 2 wt% on the electrode surface.

complex processes, due to there being less active groups on the surface [27]. Until now, therefore, these unaddressed issues making such 2D inorganic materials has not received much attention in PE for LMBs.

Inspired by aforementioned important advances in dendrite resistance design while simultaneously addressing trade-off and agglomerated effects by introducing network structures and inorganic fillers to PEs, respectively, herein we report a new class of crosslinked composite polymer electrolytes (CCPEs) consist of UV (ultraviolet)-cured vinyl IL/ethylene glycol dimethacrylate (EGDMA) (acting as the crosslinked matrix), homogeneously dispersed 2D BNNS (as the nanofiller) and IL-based electrolyte (as ion sources). The rapid and solvent-free curing of coating CCPEs can be formed on the electrodes *via in-situ* crosslinking of a precursor solution in one step (Fig. 1), not only simplifying the assembly process of LMBs but also enhancing interfacial contact/adhesion with the electrodes. IL, owing to high thermal stability and negligible vapor pressure as well as strong interaction with BNNS [28,29], is not only utilized to exfoliate bulk BN in the liquid-phase, non-covalently functionalize BNNS and assist exfoliated BNNS to homogeneously disperse in curable vinyl-IL/EGDMA monomers, also used as an electrolyte in the resulting CCPEs. Well-dispersed 2D BNNS and network structures simultaneously present in the CCPEs are found to construct 3D ion conducting paths, promote Li-ion transference number and provide IL holding ability while enhancing ionic conductivity without sacrificing mechanical strength. Moreover, impermeable BNNS enable formed a mechanically strong and highly tortuous interface to inhibit Li dendrite growth during charge-discharge cycles. The $\text{LiFePO}_4/\text{Li}$ cell assemblies with such a CCPE incorporated with 2 wt% BNNS delivers a much higher discharge capacity of 114.4 mAh g^{-1} after 200 cycles at 0.2 C and 60°C as compared to a pristine CPE (69.5 mAh g^{-1}) and CCPE with 2 wt% BN (75.1 mAh g^{-1}). The strategy has potential to make cost-effective CCPEs with excellent performances, which is envisioned to be used in the fabrication of other electrochemical systems.

2. Experimental

2.1. Materials

Bulk BN powder (99.9 wt % BN purity, 2.25 g cm^{-3} density, 3–5 μm average size of the particle) was supplied by Qingzhou Materials Co., China. ILs of 1-butyl-3-methylimidazolium hexafluorophosphate ([Bmim][PF₆]) were obtained from Henan Lihua Pharmaceutical Co., Ltd. (China) (99% purity, water content determined by KarlFischer titration: 80 ppm). 2-(dimethylamino)ethyl methacrylate (DMA, 99%,

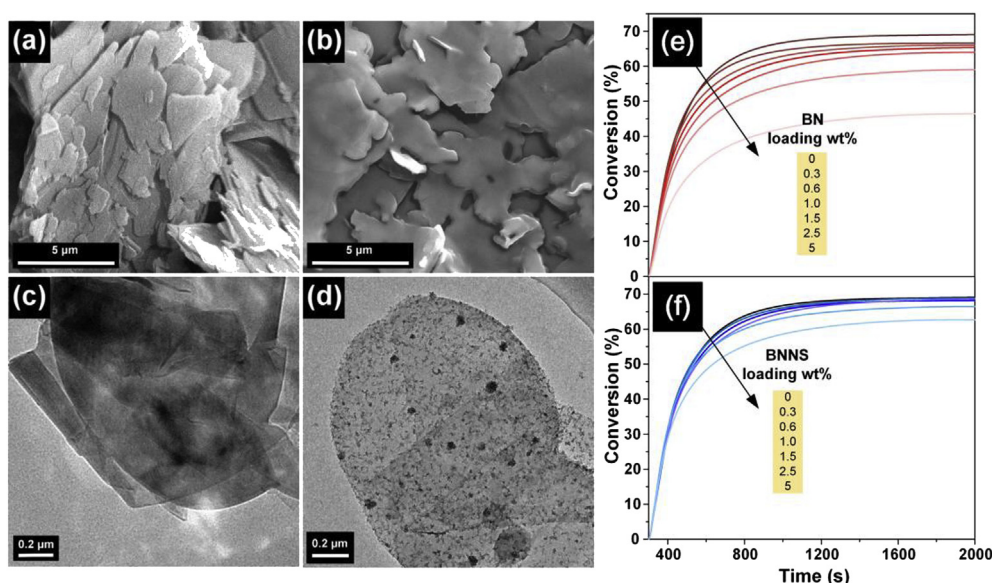


Fig. 2. SEM images of (a) BN and (b) BNNS; TEM images of (c) BN and (d) BNNS; percentage conversion profiles for the photo-crosslinking of curable PE precursor incorporated with (e) BN and (f) BNNS at 0–5 wt% filler loadings.

Acros), N-bromobutane (99%, Acros), ethylene glycol dimethacrylate (EGDMA, $\geq 99\%$, Aladdin), 2,4,6-Trimethylbenzoyldi-Phenylphosphinioxid (TPO, $\geq 99\%$, Aladdin), 1-vinyl imidazole ($\geq 98\%$, Alfa), bromobutane ($\geq 98\%$, Alfa), lithium bis(trifluoro methylsulfonyl) imide (LiTFSI, 98 + %, Admas), acetonitrile (CH_3CN , $\geq 99\%$, Aladdin), N-methyl-2-pyrrolidone (NMP, $\geq 99\%$, Aladdin), ethyl acetate (EA, $\geq 99\%$, Aladdin), anhydrous ethyl alcohol (EtOH, $\geq 99.8\%$, Sigma-Aldrich), Li foil (1 mm thickness, Aldrich), LiFePO_4 (Advanced Lithium Electrochemistry Co. Ltd., Taiwan), carbon black (CB, Ketjen black EC-600JD, Akzo Nobel) were used as received. 1-butyl-3-methylimidazolium bis(trifluoro methylsulfonyl) imide ([Bmim][TFSI]) ($\geq 99\%$) prepared according to a process previously reported in the literature (water content determined by KarlFischer titration: 100 ppm) [30].

2.2. Synthesis of 1-vinyl-3-butylimidazolium bis(trifluoro methylsulfonyl) imide (vinyl IL; [Bvim][TFSI])

In 50 ml round-bottomed flask provided with a mechanical stirrer is placed 1-vinyl imidazole (9.4 g), and bromobutane (9.8 g) is added with mild stirring at room temperature. After 2 days, the product of 1-vinyl-3-butylimidazolium bromide ([Bvim][Br]) is obtained with good yield by filtering separation. The [Bvim][Br] was then treated with equimolar LiTFSI in deionized water and stirred overnight. The final product vinyl-IL ([Bvim][TFSI]) was collected by filtration and dried at room temperature under vacuum for 12 h.

2.3. Synthesis of IL-functionalized boron nitride nanosheet (BNNS) [29]

The IL (60 ml) of [Bmim][PF₆] was added into a flask containing dried BN (300 mg), the mixture was subjected to bath sonication using an ultrasonic cleaner (B-220, 400 W; Branson Ultrasonics Corp.) for 8 h. The resulting dispersion was centrifuged at 3000 rpm for 20 min, and the supernatant containing BNNS was collected, then the BNNS/IL complexes were obtained using a vacuum-filtration. The amount of IL in the BNNS complexes was calculated using the weight loss from the TGA.

2.4. Preparation of crosslinked composite polymer electrolytes (CCPEs)

Polymeric ionic liquid (PIL)/IL/BN and PIL/IL/BNNS CCPEs with different weight proportions (see Table S1) were prepared as follows.

BN or BNNS was directly dispersed in [Bmim][TFSI] containing LiTFSI (0.5 mol kg^{-1} in IL) with ultrasonic assistance for 30 min in the absence of an organic solvent. Then, [Bvim][TFSI], EGDMA and TPO were proportionately and orderly added to the well-dispersed BNNS with warm stirring for 1 h, the precursor of CCPE was obtained. The amount of BN or BNNS was controlled as 0.3, 0.6, 1.0, 1.5, 2.0 wt% of the curable monomers including [Bvim][TFSI] and EGDMA (accounted for 10% of [Bvim][TFSI]). The precursor was cast onto the glass plate, and then exposed to UV irradiation for 60 s, leading to the free-standing film with a thickness of $\sim 100 \mu\text{m}$. The pure PIL/IL CPE was prepared by a similar procedure with weight proportions of IL, vinyl-IL, EGDMA and TPO at 40, 53, 6, 1%, respectively.

3. Results and discussion

3.1. Structural and morphological character of crosslinked composite polymer electrolytes (CCPEs)

In this study, BN can be exfoliated and non-covalently functionalized simultaneously by adding IL of [Bmim][PF₆] into BN with mild batch sonication due to the presence of strong cation- π interactions between IL cations and BNNS and then removal of un-exfoliated BN via centrifugation step [29,31]. After removal of most of free IL by vacuum filter, $\sim 64 \text{ wt\%}$ of IL was estimated by TGA result (Fig. S1), which value was higher than the BNNS non-covalently functionalized with IL in our previous study [31] ($\sim 17 \text{ wt\%}$) indicating that the resulting BNNS/IL complexes consisted of free IL and physically absorbed IL. The morphologies of the obtained BNNS were further investigated by scanning electron microscopy (SEM) and transmission electron microscope (TEM) [Fig. 2(a-d)]. It can be seen that the sheet-shaped morphology of BN and BNNS was similar in their SEM images, while edge of BNNS was blurrier than that of BN due to the presence of IL on BNNS surface. As shown by TEM in Fig. 2(c)-2(d), a few layers of BNNS with surface immobilized IL were observed revealing that BN was exfoliated into BNNS and its surface was homogeneously decorated by IL. The obtained BNNS with IL assistant can be dispersed in ILs as well as in various organic solvents, which is ready to be available for CCPEs fabrication.

It is well known there are two typical strategies for preparing IL-based PES, such as (i) doping of polymer with ILs and (ii) polymerization/crosslinking of monomers in ILs, which are both candidates for

consideration in this study [14]. For the former approach, although surface immobilized IL afforded a stable BNNS suspension in various organic solvents, however, BNNS starts to sedimentation in PIL/IL/CH₃CN solution after few hours as shown in Fig. 1. We speculated that surface immobilized IL might partially remove from BNNS surface in CH₃CN during the physical blending, which may deteriorate the BNNS dispersion in the matrix during the solvent evaporation process. In contrary, the obtained BNNS can be quickly dispersed in the curable CPE precursor to form homogenous suspension at high concentration ($> 20 \text{ mg mL}^{-1}$) by a simple blending. Therefore, the CCPEs can be made with ready-to-use curable precursor containing a high content of well-dispersed BNNS by photocrosslinking process.

Before the photocrosslinking process, it is crucial to investigate how these fillers influence the curing behavior within the matrix, because of the presence of residual uncured monomers are highly reactive with Li metal, which results in the increasing electrode resistance thereby severely weakening the battery performance. For this propose, photo-differential scanning calorimeter (DSC) was employed to clarify the photocuring process of the UV-curable precursor containing different filler loadings (from 0 to 5 wt%) of BNNS (the proportion as shown in Table S1). For a comparison, incorporated BN precursor systems were also examined, and the “conversion-time” kinetics curves for both CCPEs were shown in Fig. 2(e) and (f). It was observed that the ultimate percentage conversion of precursor with BN exhibited lower cross-linking reactivity than for the precursor with BNNS under similar filler loadings, which was probably attributable to BN with micrometer size as a barrier for the photocrosslinking process, thereby hindering the absorption of the incident radiation by the photoinitiator. Noteworthy, only a slight decrease and increase in the percentage conversion and the induced time when BNNS loading more than 2.5 wt%, respectively. To ensure high photoinitiation efficiency of UV curing for the BNNS system, a maximum filler loading of 2 wt% in the PE precursor were used in this study and their curing process was further examined by FTIR spectra (Fig. S2). The FTIR spectrum of vinyl-IL and EGDMA represent the characteristic peaks at 1466 and 1661 cm^{-1} corresponding to reactive C=C double bonds and 1639 cm^{-1} corresponding to acrylic C=C double bonds, respectively, which have almost disappeared after UV curing in CPE or CCPEs. These results suggest that the crosslinking of PE precursors with fillers were realized after UV curing.

Fig. 3(a)–(d) show the SEM images of CCPEs with BN and BNNS at 1.0 and 2.0 wt% filler loadings. An apparent agglomeration of the BN plates with micro-sized scale is embedded in both PIL/IL/BN CCPEs. By contrast, improved dispersion and interfacial bonding by surface IL functionalized of BNNS is observed, and no visible agglomeration and layers stacking of BNNS are apparent evidenced by the observation of the flaky BNNS bulge in the resulting CCPEs. However, it was noted that the dispersion degree of PIL/IL/BNNS CCPEs prepared by doping of PIL with IL (Fig. S3) is entirely different from that found by UV cross-linking of incorporated BNNS PE precursor. That is, visible BNNS agglomeration and severely stacking even at a low filler loading of 1 wt% for the PIL/IL/BNNS prepared by doping of PIL with IL, which is in a good agreement with the previous observation by its unstable dispersion in CH₃CN. This result indicates that immobilized IL on exfoliated BNNS surface not only assists BNNS dispersion in PE precursor before UV curing, but also prevents BNNS stacking during UV curing process. Finally, it becomes a part of the CCPE to strengthen the BNNS/PE interfacial interaction and facilitates the Li-ion transfer in the resulting CCPE simultaneously. Moreover, the homogeneously dispersed 2D BNNS in the resulting CCPEs is propitious to construct the continuous and interconnected ion transfer pathways for the optimization of ion transport properties.

3.2. Electrochemical properties of crosslinked composite polymer electrolytes

Before characterizing the ionic conductivity of obtained CCPEs, the

effect of both fillers in the IL-based electrolyte (containing LiTFSI at concentration of 0.5 mol kg^{-1} in IL) was elucidated by measuring the characteristic viscosity and ionic conductivity of BN/IL or BNNS/IL mixtures at different filler concentrations (1, 3 and 5 mg mL^{-1}) (Table S2). Fig. 4(a) shows that the viscosities for both mixtures increase with increasing filler content at all temperatures, while the more noticeable change of viscosities is observed in the incorporated BNNS mixtures. This result is consistent with other introduced filler IL systems with improved dispersion and interfacial bonding from surface filler functionalization [23,32]. In addition, the viscosity activation energy (E_{av}) calculated by Vogel-Tammann-Fulcher (VTF) equation (see supporting information) of pure IL ($10.38 \times 10^{-21} \text{ J}$) is decreased with increasing filler content, it decreased to 8.53×10^{-21} and $9.23 \times 10^{-21} \text{ J}$ as the filler content of BN and BNNS in the IL reaches 5 mg mL^{-1} , respectively [Fig. 4(c)]. We speculated that the presence of strong cation- π interaction between BN/or BNNS and IL cations leading to a reduction of interactions between IL molecules, thereby increasing IL molecules mobility and lowering E_{av} value of resultant mixtures. The ionic conductivities and their activation energy (E_{ao}) calculated by VTF equation of the IL, BN/IL and BNNS/IL mixtures were also examined and shown in Fig. 4(b) and (d), respectively. As expected, by adding both BN or BNNS, E_{ao} of IL ($9.90 \times 10^{-21} \text{ J}$) decreased significantly and reached 8.18×10^{-21} and $7.26 \times 10^{-21} \text{ J}$ when filler content was 3 mg mL^{-1} , respectively. However, further addition of fillers slightly increases the E_{ao} value, which is not consistent with above experimental viscosity observations, indicating that further added filler has a relatively weak promoting effect on ionic conduction. Also, it is worth noting that the improvements in ionic conductivities of IL are more evident by adding BNNS compared to by adding BN, highlighting that BNNS has more effective in ion transport caused by its better dispersion and larger contact areas between nanostructured BNNS and IL molecules.

Fig. 5(a) and (b) show the through-plane ionic conductivities of CCPEs containing BN and BNNS plotted as a function of temperature (30–80 °C). It is clear that both 2D fillers facilitate the formation of ion-conducting channels in resulting CCPEs leading to a visible improvement in ionic conductivity, especially introducing BNNS. Maximum ionic conductivity, that is, $2.7 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C is obtained with BNNS 1.5 wt% loading in PIL/IL CPE, which is twice more than pure CPE. These results are in a good agreement with previous SEM observation, the filler with the nanostructure and surface functionalization show greater dispersibility in and compatibility with CPE matrix, forming more IL channels and ion carriers thereby enhancing the ion transport more effectively. However, such improvement starts to weaken slightly by further introducing fillers in the CCPEs at 2.0 wt%, which may result from barrier effect on the conducting channels caused by excess 2D fillers [23,33]. This influence is also reflected in the E_a of the CCPEs [Fig. 5(c)], which show that calculated E_a values for both CCPEs significantly increase at high filler content. When compared with the calculated E_a values of both CCPEs at different filler contents, it is clear that the ion transfer process needs less energy in the PIL/IL/BNNS CCPEs. This indicates that the presence of BNNS may introduce additional conducting channels as well as strength ionic conduction at BNNS/CPE interface, thereby increasing low-energy pathways for ion transfer and lowering E_a in the resulting CCPEs.

After characterizing the ionic conductivity of obtained CCPEs, their Li-ion transfer number (t_{Li+}) measurements were further performed to characterize the CPE and CCPEs with an optimum filler loading of 1.5 wt% for battery applications by recording AC impedance and chronoamperometry (see supporting information; Fig. S4). The resultant t_{Li+} value evaluated at 60 °C are 0.38, 0.59 and 0.67 for crosslinked PIL/IL PE, PIL/IL/BN 1.5 wt% and PIL/IL/BNNS 1.5 wt% CCPEs, respectively. Significantly, the combined dispersion, interfacial interaction and ion mobility enhancements appear to be effective in giving overall increases of ~ 1.8 -fold in t_{Li+} . Besides t_{Li+} , another important property for evaluating these CCPEs as candidates for Li-metal batteries, that is the electrochemical stability as evaluated by

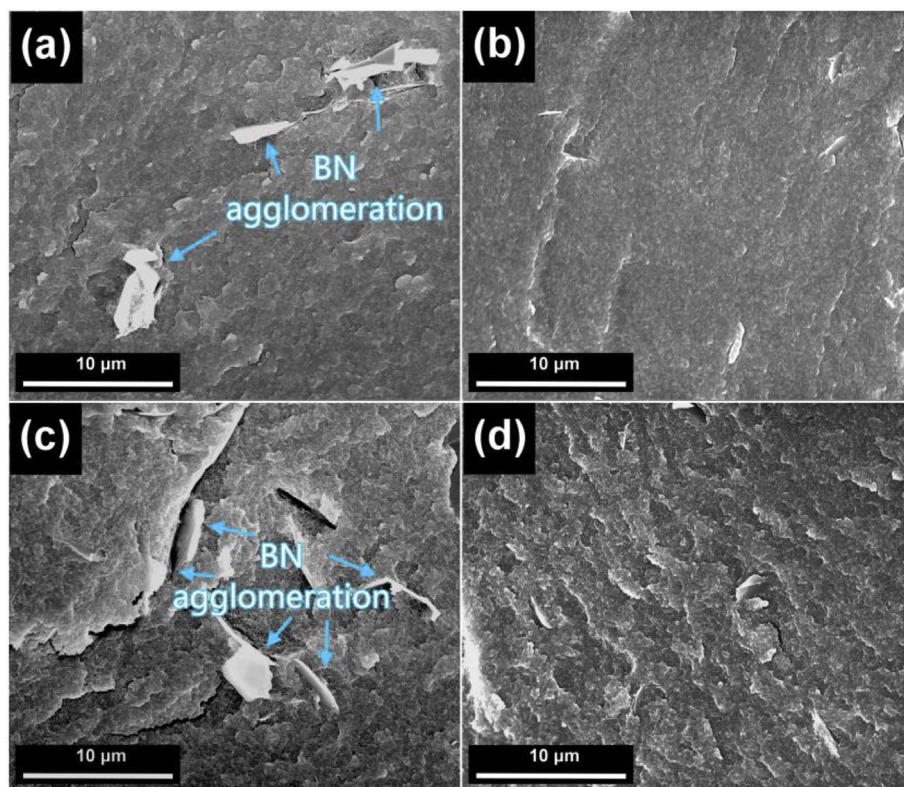


Fig. 3. SEM images of fracture surfaces of (a) PIL/IL/BN 1.0 wt%, (b) PIL/IL/BNNS 1.0 wt%, (c) PIL/IL/BN 2.0 wt% and (d) PIL/IL/BNNS 2.0 wt%.

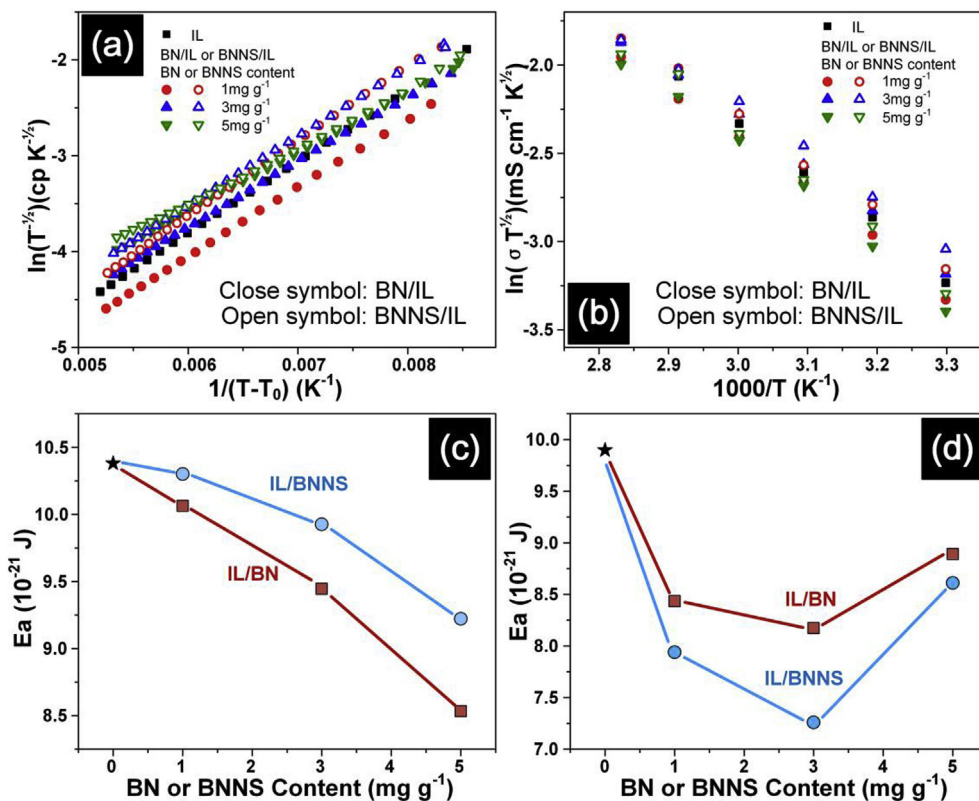


Fig. 4. VFT plot of (a) viscosities and (b) conductivities versus temperature of IL, BN/IL and BNNS/IL mixtures with filler contents; Variation of (c) $E_{a\eta}$ and (d) $E_{a\sigma}$ of IL and BN/IL and BNNS/IL mixtures with filler contents.

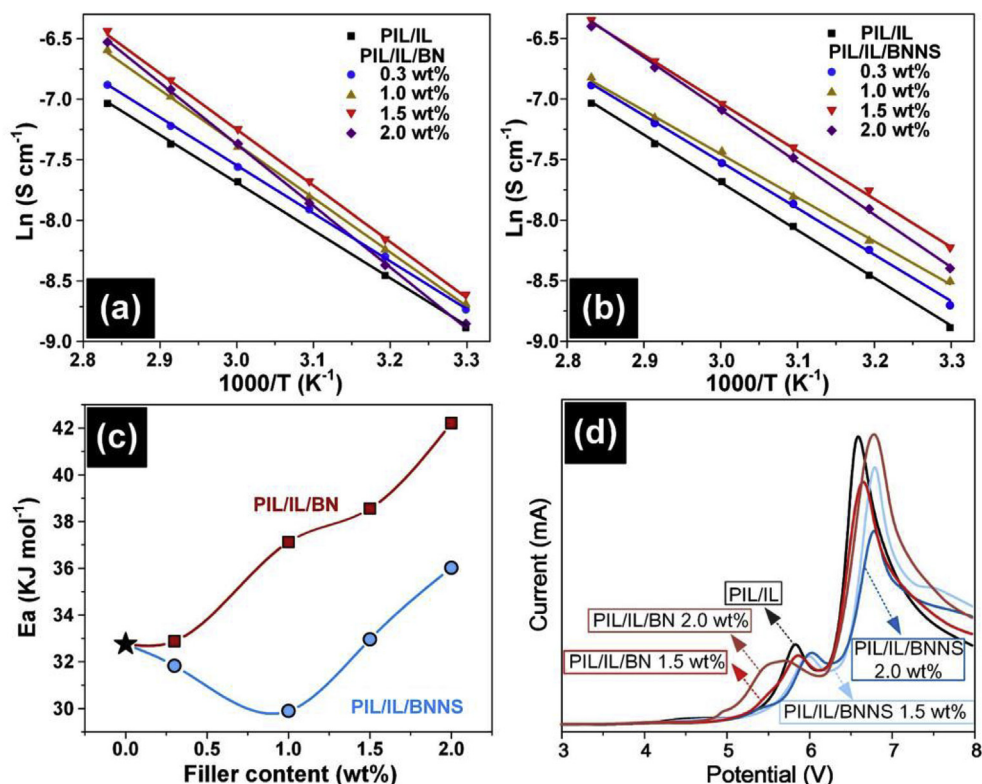


Fig. 5. (a) and (b) Temperature dependence of ionic conductivity for crosslinked PIL/IL PE, PIL/IL/BN and PIL/IL/BNNS CCPEs; (c) Variations of E_a of PIL/IL/BN and PIL/IL/BNNS CCPEs with filler content; (d) LSV profile of PIL/IL PE, PIL/IL/BN and PIL/IL/BNNS CCPEs at 60 °C.

their linear sweep voltammetry (LSV) curves at 60 °C. From Fig. 5(d), a wide electrochemical stability window (ESW) for plain CPE can be found below 5.2 V (vs. Li/Li^+), which is more stable than previous reported PIL/IL PE (unstable oxidation peak in the voltage higher than 4.2 V vs. Li/Li^+) [34,35], which can be described by the introduction of crosslinked structure in PIL-based PE. Owing to well-dispersed exfoliated BNNS possessing large specific areas and strong interfacial interaction, the anodic limiting potential of the introduced BNNS CCPEs was about 5.5 V (vs. Li/Li^+). By contrast, agglomerated BN fail to provide a robust interface thereby resulting in poor electrochemical stability.

To further investigate the effect of BN and BNNS on the interfacial stability with Li metal, we assessed the symmetrical cell of $\text{Li}|\text{electrolyte}|\text{Li}$ containing the CPE and CCPEs after different storage days at 60 °C. As shown in Fig. S5(a)–S5(c), the electrolyte bulk resistance (R_B), assigned by the intercept with real axis of the response at high frequency, for the cell with PIL/IL or PIL/IL/BN 2.0 wt% was slightly increased with time during 36 days at 60 °C, indicating the ionic conductivity of these CPE or CCPE undergoes decline during long-term storage. Also, the overall interfacial resistance (R_I) of electrolyte/Li metal, represented by the diameter of capacitive arcs in AC impedance spectra, slightly increase with rising of storage time and tends to a stable value gradually after 9 days, leveling at about 890 and 972 Ω for the cells with PIL/IL and PIL/IL/BN 2.0 wt% after 36 storage days at 60 °C, respectively. Whereas in the presence of BNNS, the R_B and R_I of the resulting cell slightly increase during the initial period of storage and tends to a stable value after 5 days, leveling about 408 Ω for R_I after 36 storage days at 60 °C. Li redox in the CPE and CCPEs was characterized by cyclic voltammograms (CVs) with $\text{Li}|\text{electrolyte}|\text{Li}$ cell, as shown in Fig. S5(d)–S5(f). In the initial cycle for CPE, the anodic oxidation at +0.33 and +1.10 V and a cathodic reduction at -0.39 ~ -0.06 V vs. Li/Li^+ , which show a visible change in the following cycles. A similar result was also collected for the CCPE with 2.0 wt% BN. However, the incorporated BNNS 2.0 wt% CCPE afforded a slighter

change in the charge transfer showing nearly overlapping CV curves after the third cycle. These results demonstrating that the compatibility of the introduced BNNS CCPE with Li electrodes is better than pure PIL/IL and PIL/IL/BN. It can be attributed to that well-dispersed BNNS the in favor of forming a stable SEI layer on Li metal and preventing further reaction of an organic matrix with Li metal [36].

3.3. Electrochemical performance of cells assembled with crosslinked composite polymer electrolytes

It is worth noting that CCPEs containing BNNS can form on the LiFePO_4 electrodes by rapid and solvent-free UV-curing of their precursor solution in one step. As shown in Fig. S6, the SEM image of LiFePO_4 electrode coated with the PIL/IL/BNNS 2.0 wt% CCPE. Its fracture surface shows that the resultant CCPE with ~30 μm thickness is firmly adhesive with a LiFePO_4 electrode without any defects. Such in-situ prepared CCPEs containing well-dispersed BNNS can construct a robust electrode/electrolyte integrated structure with enhanced interfacial contact/adhesion as well as low interface resistance as further discussed in the following section. By assembling the CPE or CCPEs coated LiFePO_4 electrode with Li metal, the assembly process of $\text{LiFePO}_4/\text{Li}$ cells were simplified and obtained cells were utilized to investigate the electrochemical performance of CPE or CCPEs in Li metal batteries. Fig. 6(a)–(c) displays the cycle performances of cells with CPE and CCPEs at a constant charge-discharge current of 0.2 C and 60 °C. The cell with pure CPE reveals an initial discharge capacity of 140.5 mAh g^{-1} and a capacity retention ratio of 49.5% after 200 discharge-discharge cycles. Introducing BN in the CPE matrix show only a limited improvement in cell performance, the resulting cells exhibit initial discharge capacity of 145.1–153.3 mAh g^{-1} and capacity retention ratio of 47.2–50.0% after 200 discharge-discharge cycles. By contrast, both initial discharge capacity and capacity retention ratio of the cell are significantly improved by the assembling introduced BNNS CCPEs. It is noted that the initial discharge capacity (150.0 mAh g^{-1}) and

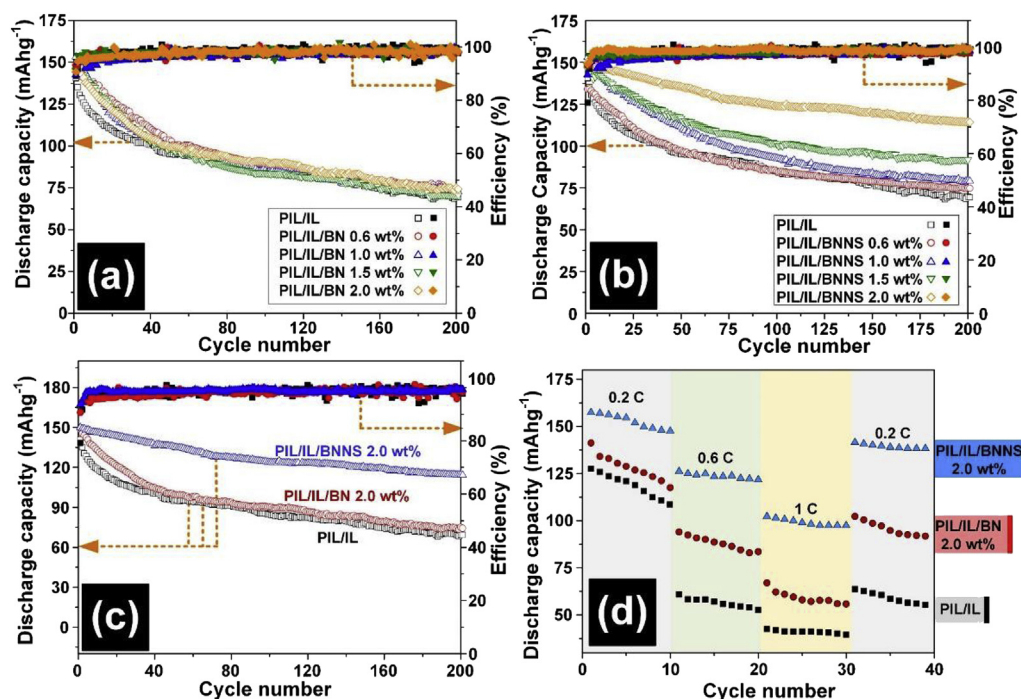


Fig. 6. (a)–(c) Charge-discharge capacity and coulombic efficiency as a function of cycle number for LiFePO₄/Li cells with the PIL/IL CPE, PIL/IL/BN and PIL/IL/BNNS CCPEs at 60 °C with 0.2 C current rate; (d) Rate capabilities of LiFePO₄/Li cells with the PIL/IL CPE, PIL/IL/BN and PIL/IL/BNNS CCPEs at 60 °C and different current densities.

capacity retention ratio (76.2%) of the cell with PIL/IL/BNNS reaches a maximum when the BNNS are loading up to 2.0 wt% after 200 discharge-discharge cycles. This result suggests that more BNNS loading in the crosslinked matrix under same conductivity level ($> 8 \times 10^{-4} \text{ S cm}^{-1}$) has a greater ability to deliver a relatively stable and robust interface during charge-discharge cycles.

Fig. 6(c) further reveals the comparison of discharge capacity and efficiency for the cells with pure CPE and CCPEs at 2 wt% filler loading. The Coulombic efficiency of the cell with PIL/IL/BNNS achieves nearly 100% coulombic efficiency during the long-term cycles, maintaining 99.4% at the end of 200 cycles. Moreover, it displays a higher capacity (114.4 mAh g⁻¹), with $\sim 65\%$ enhancement, compared to the cell with pure CPE after 200 cycles at 60 °C (69.5 mAh g⁻¹). However, the cell with PIL/IL/BN only keeps the Coulombic efficiency of 98.0% and limited enhancement of $\sim 8\%$ after 200 cycles. This result indicates that well-dispersed BNNS more effectively improved interfacial compatibility of the electrolyte with CPE matrix compared to agglomerated BN. As compared with the cell performance of other reported PIL-based PE (Table S3), the present system with PIL/IL/BNNS CCPE shows better, or comparable cycling stability and capacity retention than these reported earlier under similar test conditions. The rate performance of cells with CPE and CCPEs at various rates from 0.2 to 1 C is also investigated, and the results are shown in Fig. 6(d). When increasing the discharge rate to 0.6 and 1 C, the discharge capacity of the cell with PIL/IL significantly diminish to 60.9 and 20.0 mAh g⁻¹, respectively, which results from the increase in cell polarization effect [37]. Moreover, its specific capacity cannot be restored (31.1 mAh g⁻¹) as the current density turns back to 0.2 C. However, for the cell with PIL/IL/BNNS 2.0 wt%, acceptable discharge capacities of 126.4 and 104.5 mAh g⁻¹ are achieved at 0.6 and 1 C, respectively. Moreover, the cell can achieve stable discharge capacity upon cycling at each current rate and restore a high discharge capacity of the 142.5 mAh g⁻¹ when the rate returned to 0.2 C. Whereas for the cell with PIL/IL/BN 2.0 wt%, raising the discharge rate still leads to an apparent drop the discharge capacity at each current rate and shows a limited improvement in discharge capacity when the rate returned to 0.2 C (102.9 mAh g⁻¹). These results are demonstrating improved electrochemical performance and high reversibility of CCPE with BNNS.

To gain further understanding about the effect of BNNS on cycling

stability, we compare the charge-discharge curves [Fig. 7(a)–(c)] and corresponding electrochemical impedance spectroscopy (EIS) before and after cycling test [Fig. 7(d)–(f)]. As shown in Fig. 7(c), it is noted that the cell using BNNS 2.0 wt% depicts a voltage plateaus at first cycle, and then shows a slight change of the potential gap with cycling (from 0.13 to 0.28 V). The EIS measurement result shows the corresponding charge-transfer resistance (R_{CT}) value, represented by a semicircle at the medium frequency, first slight increases to 231 Ω after 3 cycles and remains constant during the subsequent cycling [Fig. 7(f)]. However, for the pure PIL/IL and PIL/IL/BN 2.0 wt%, a noticeable change of discharge plateau voltage and potential gap (PIL/IL: 0.29–0.89 V; PIL/IL/BN: 0.10–0.60 V) is observed with cycling [Fig. S7(a)]. The corresponding R_{CT} values of the PIL/IL and PIL/IL/BN 2.0 wt% rose very fast during the cycling and reached 2142 and 1000 Ω after 150 cycles, respectively, which are much higher than that of CCPE with BNNS ($R_{CT} = 397 \Omega$) [Fig. S7(b)]. This phenomenon indicates that the introduction of BNNS improved the interfacial properties and kept relatively stability and robust contact between the CPE and electrode during charge-discharge cycles.

To further clarify the function of BNNS on the interfacial stability and charge transfer behavior, the Li/Li symmetric cells with PIL/IL and PIL/IL/BNNS 2.0 wt% were employed and their voltage-time profiles were shown in Fig. 7(g). It is seen that the overpotential for deposition or stripping of Li metal in the cell with PIL/IL CPE at a current density of 0.1 mA cm⁻² is about 20 mV at the beginning and starts to increase quickly after 90 h. After 130 h, the symmetric cell exhibits a large and irreversible voltage drop, which can be attributed to the short-circuiting behavior in the cell by dendrite growth [2,38]. In comparison, the cell with PIL/IL/BNNS 2.0 wt% maintains a relatively low (~ 10 mV) and stable voltage polarization without short-circuiting for over 200 h under the same conditions. Increasing the current density to 0.2 mA cm⁻² causes the higher overpotential for deposition or stripping of Li metal in both cells at the beginning. It is worth to note that the symmetric cell with PIL/IL/BNNS 2.0 wt% CCPE exhibits much stable voltage polarization as compared to the cell with PIL/IL CPE, which maintain about 20 mV of overpotential for deposition or stripping of Li metal after 200 h. It is demonstrated that the PIL/IL/BNNS CCPE exhibits a more stable Li deposition/stripping cycling reversibility and a better ability for restraining the Li dendrite growth as compared to the

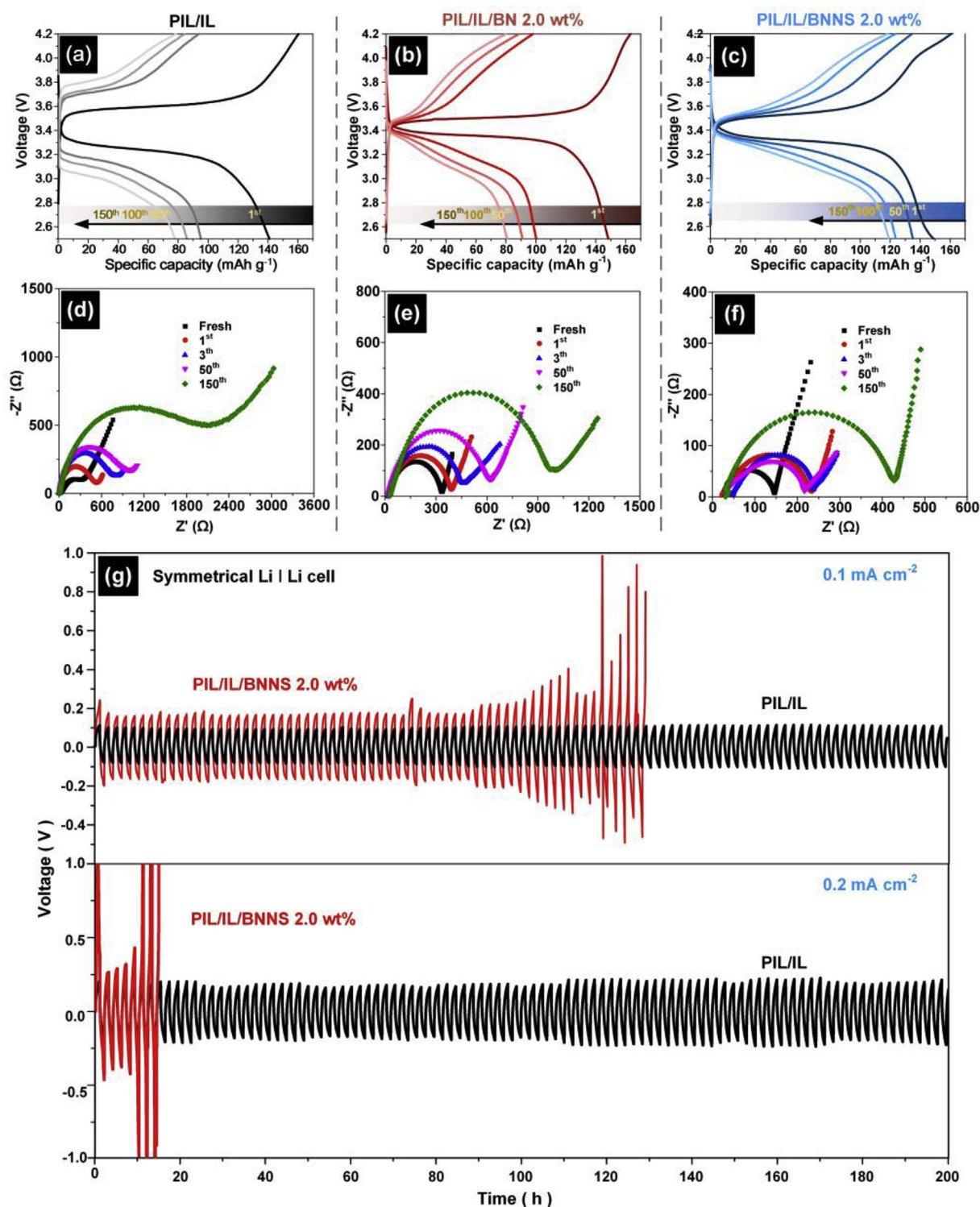


Fig. 7. (a)–(c) Charge-discharge profiles and (d)–(f) EISs of cells with PIL/IL CPE, PIL/IL/BN 2.0 wt% and PIL/IL/BNNS CCPEs at 60 °C with 0.2 C current rate; (g) Voltage-time profile of the symmetrical Li|Li cells with PIL/IL CPE and PIL/IL/BNNS CCPEs at 60 °C. The current density was fixed at 0.1 and 0.2 mA cm⁻² with a stripping/plating capacity of 0.1 and 0.2 mAh cm⁻², respectively.

PIL/IL CPE. Therefore, such excellent compatibility between the Li metal electrode and CCPE ensures a long life for the fabricated batteries.

Fig. 8 and Fig. S8 show the surface morphologies of the pristine Li anode and Li anodes harvested from the cycled LiFePO₄/Li cells after 150 cycles at 60 °C with 0.2 C current rate, respectively. Compared with the smooth and dense surface of pristine Li anode before cycling,

irregular Li dendrites are visible on the surface of Li anode disassembled from the cell with pure PIL/IL PE after 150 cycles at 0.2 C [Fig. 8(a) and (b)]. However, for cell based on introduced BNNS CCPE, the surface of the Li anode remained relative smooth after 150 cycles, as shown in Fig. 8(d) and (e). Furthermore, the SEM images of the CPE with and without BNNS adjacent to Li anode harvested from the cycled Li metal cells were further to verify the suppression of 2D BNNS against Li

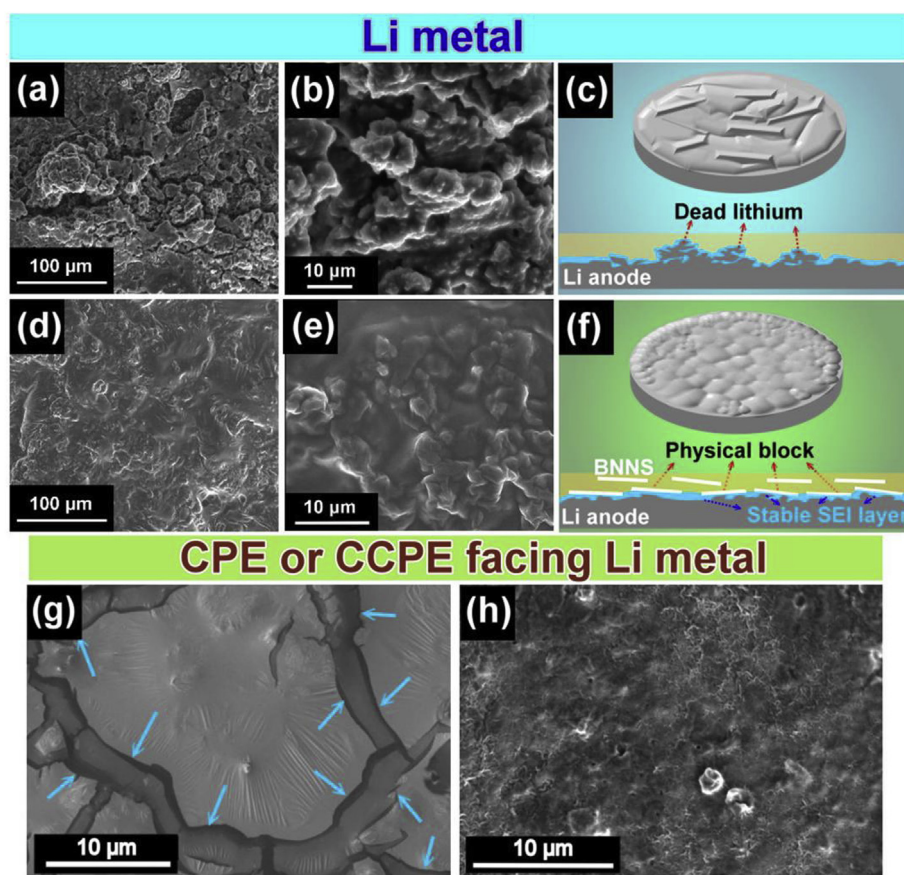


Fig. 8. Schematic diagram and SEM images of the surface of the Li electrodes obtained from the cells with (a)–(c) PIL/IL CPE and (d)–(f) PIL/IL/BNNS 2.0 wt% CCPE after 150 cycles at 60 °C with 0.2 C current rate; SEM images of (g) PIL/IL CPE and (h) PIL/IL/BNNS 2.0 wt% CCPE after 150 cycles at 60 °C with 0.2 C current rate.

dendrite growth. From Fig. 8(g), there are apparent damages caused by Li dendrites on the surface of PIL/IL CPE, whereas the PIL/IL with BNNS [Fig. 8(h)] presents a flat surface with a small number of bulges after 150 cycles. These results provide good evidence that introduced well-dispersed 2D BNNS in the CPE matrix can inhibit the Li dendrite growth during charge-discharge cycles, which can be illustrated by the following reasons [Fig. 8(c) and (f)]: (i) Homogeneously dispersed BNNS in CCPE facilitate the construction of continuous and interconnected ion-conducting paths for reducing charge concentration gradient in the electrolyte during Li plating and stripping. (ii) Introduced BNNS with high Young's modulus provides a mechanically robust blocking against the Li dendrites formation and growth, resulting in a stable SEI with low resistance [21]. Therefore, the combination of high Li-ion conductivity and stable SEI appears to be a reason for LMBs with good cycling and rate performance.

4. Conclusions

In summary, the strategy of UV assisted *in-situ* crosslinking was employed to fabricate IL-based CCPE with homogeneously dispersed BNNS, which provides a practical and straightforward pathway to assemble LMBs. Moreover, it has been shown that the presence of both network structure and 2D BNNS in PE matrix is possible to create a facile channel for Li⁺ transport and an organic-inorganic skeleton for strengthening IL holding ability and interface strength. The results demonstrated that the LiFePO₄/Li cell assembled with incorporated 2.0 wt % BNNS CCPE is favorable to constitute the stable and robust interface, and consequently achieve excellent electrochemical performance. The resultant cell present high-performance charge-discharge behavior with a high discharge capacity of 114.4 mAh g⁻¹ after 200 cycles at 0.2 C

and adequate discharge capacity of 104.5 mAh g⁻¹ at 1 C and 60 °C. This facile obtained CCPE in combination with its excellent battery performance exerted a beneficial influence on the development of high-performance LMBs.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2019.01.004>.

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